

Experiment – 5

Analysis of Data Communication in a Web Application Using Browser Development Platform

Aim

To analyse how data is communicated between a web browser and a web server using the **Developer Tools** available in a browser.

Objective

- To understand browser-server communication.
- To observe HTTP/HTTPS requests and responses.
- To analyse request and response headers.
- To identify HTTP methods and status codes.
- To observe the data transferred between the browser and server.

Requirements

- Computer/Laptop
- Internet Connection
- Google Chrome / Microsoft Edge
- Any web application
- Browser Developer Tools

1. Introduction

A web application communicates with a server whenever the user loads a page, submits a form, or requests some information. The browser sends a request to the server, and the server processes it and returns a response.

The browser's **Developer Tools** provide a simple way to observe this communication. The **Network** tab can be used to view the request URL, HTTP method, status code, headers, response data, and loading time.

Basic Communication Flow



2. Procedure

Step 1 – Open the Web Application

- Open Google Chrome or Microsoft Edge.
- Open any web application in the browser.

Step 2 – Open Developer Tools

- Press **F12** or right-click the webpage and select **Inspect**.
- Open the **Network** tab.

Step 3 – Capture Communication

- Reload the webpage.
- Observe the requests generated by the browser.
- Select any request from the Network list.

Step 4 – Analyse the Request

Check the following details:

- Request URL
- Request Method
- Request Headers
- Query Parameters
- Request Payload, if available

Step 5 – Analyse the Response

Check:

- Status Code
- Response Headers
- Response Data
- Content Type
- Response Time

Step 6 – Record the Observation

Record the important communication details in an observation table.

3. Observation and Analysis

Parameter	Observation
Request Method	GET / POST
Protocol	HTTP / HTTPS
Status Code	200 – Successful
Request URL	Web application/API URL
Content Type	HTML / JSON
Response	Webpage or requested data
Response Time	Displayed in Network tab
Data Size	Displayed in Network tab

Common HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Send data to server
PUT	Update existing data
DELETE	Remove data

Common Status Codes

Status Code	Meaning
200	Request successful
201	Resource created
400	Bad request
401	Unauthorized
404	Resource not found
500	Server error

4. Request and Response Analysis

When a user opens a web application, the browser sends an HTTP/HTTPS request to the required server. The request contains information such as the URL, method, headers, and sometimes user data.

The server processes the request and sends a response back to the browser. The response contains a status code, headers, and the requested content such as **HTML, JSON, images, or other resources**.

The Network tab helps us observe this complete communication process without directly accessing the server.

5. Result

The data communication between the browser and web server was successfully analysed using the browser's Developer Tools. The HTTP methods, request and response headers, status codes, content type, and data transfer details were observed through the **Network** tab.

6. Conclusion

The experiment helped in understanding how a web application communicates with its server through HTTP/HTTPS. Browser Developer Tools provide a useful platform for monitoring and debugging this communication, making it easier to identify request errors, response problems, and performance issues.